

PIPELINE FAILURE RCA REPORT

Job Name:

Nightly ETL Job

Failure Time:

2025-06-07 02:00:09

Root Cause:

Database connection failure due to timeout

Evidence:

- Failure log shows repeated timeout exceptions when attempting to connect to db-prod.internal:5432
- Success log from the previous day shows a successful database connection to the same host and port
- Recent git changes do not indicate any modifications to database connection settings or related code

Log Differences:

Missing in failure:

```
- 2025-06-06 02:00:01 [INFO] Pipeline: Nightly ETL Job started
- 2025-06-06 02:00:02 [INFO] Loading configuration from config.yaml
- 2025-06-06 02:00:03 [INFO] Database connection established (host: db-prod.internal, port: 5432)
- 2025-06-06 02:00:04 [INFO] Connection pool initialized: 10 connections, timeout=10s
- 2025-06-06 02:00:05 [INFO] Step 1/5: Extracting data from source API
- 2025-06-06 02:01:30 [INFO] Extracted 45,000 records from /api/v2/orders
- 2025-06-06 02:01:31 [INFO] Step 2/5: Validating schema
- 2025-06-06 02:01:35 [INFO] Schema validation passed: all 45,000 records valid
- 2025-06-06 02:01:36 [INFO] Step 3/5: Transforming data
- 2025-06-06 02:03:10 [INFO] Transformation complete: 45,000 records processed
- 2025-06-06 02:03:11 [INFO] Step 4/5: Loading into data warehouse
- 2025-06-06 02:04:45 [INFO] Load complete: 45,000 rows inserted into dw.fact_orders
- 2025-06-06 02:04:46 [INFO] Step 5/5: Running data quality checks
- 2025-06-06 02:05:00 [INFO] Data quality checks passed
- 2025-06-06 02:05:01 [INFO] Sending notification: success
- 2025-06-06 02:05:02 [INFO] Pipeline completed successfully in 5m 1s
```

Unexpected in failure:

```
+ 2025-06-07 02:00:01 [INFO] Pipeline: Nightly ETL Job started
+ 2025-06-07 02:00:02 [INFO] Loading configuration from config.yaml
+ 2025-06-07 02:00:03 [WARN] Database connection attempt 1/3 failed
+ 2025-06-07 02:00:03 [ERROR] TimeoutException: Connection to db-prod.internal:5432 timed out after 2000ms
+ 2025-06-07 02:00:04 [WARN] Retrying database connection (attempt 2/3)...
+ 2025-06-07 02:00:06 [ERROR] TimeoutException: Connection to db-prod.internal:5432 timed out after 2000ms
+ 2025-06-07 02:00:06 [WARN] Retrying database connection (attempt 3/3)...
+ 2025-06-07 02:00:08 [ERROR] TimeoutException: Connection to db-prod.internal:5432 timed out after 2000ms
+ 2025-06-07 02:00:08 [ERROR] Database Connection Failure: All 3 connection attempts failed
+ 2025-06-07 02:00:09 [ERROR] Pipeline aborted at Step 1: unable to establish database connection
```

+ 2025-06-07 02:00:09 [FATAL] Nightly ETL Job failed - exiting with code 1

Impact:

The Nightly ETL Job was aborted, and data was not extracted, transformed, or loaded into the data warehouse

Recommended Action:

1. Investigate network connectivity issues between the pipeline and db-prod.internal
2. Verify database server status and check for any maintenance or downtime schedules
3. Consider increasing the database connection timeout or implementing a retry mechanism with exponential backoff

Confidence Score:

80%

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